Targeted Value CpIO alignment

How we are eliminating measurement errors and ensuring accurate CPiO across all markets.

Last updated on March 2026

1. Current Campaign Evaluation Setup

Current Campaign Evaluation Setup

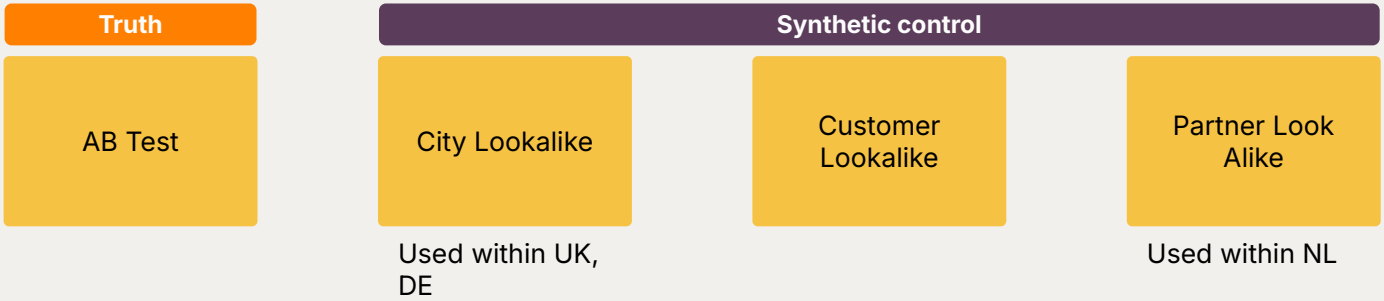
Standardizing campaign evaluation across markets

		AUDIENCE	AS IS	PROPOSAL	TESTS
1	Smart Growth	Early Engaged	Voucher Control	Voucher Control	Smart Growth (Wave 1) - DE Smart Growth (Wave 2) - DE
2	Winback	Dormant Churned Lapsed	Voucher Control	Voucher Control	Winback -UK
3	SVP	New	Voucher Control	Voucher Control	
4	Reactivate	Lapsed	City Look Alike	Customer Look Alike	
5	Local Heroes	Prospects	City Look Alike	City Look Alike	

2. Campaign Analysis

Overview of evaluation techniques - AB test vs. synthetic control by city, customer and partner

The golden standard to measure causal effects of our campaigns are unbiased A/B tests



Overview of evaluation techniques - AB test vs. synthetic control by city, customer and partner

	Treatment definition	Control definition	Considerations
AB Test	Customers that did receive email/push in focus cities	Unbiased hold-out group of customers in same cities	Golden Truth
City Lookalike	All customers of targeted segment in focus cities	Cities identical to focus cities from the same country in the last 12 months (selected using KNN)	- Significance and incrementality often change with multiple runs because of bayesian uncertainty - Control cities are fixed before Q5, may diverge over time - Overlap and double counting of other campaigns
Customer Lookalike	Customers that did receive email/push in focus cities	For each treatment customer, look alikes are found outside the focus cities in last 12 months	- Customers could have look alikes from cities with completely different dynamics - Not possible for customers without order history (new, reactivated, churned) *Next step: add geographical + density matching conditions*
Partner Look Alike	All partners in focus cities	For each partner in focus city, look alikes are found outside the focus cities in last 12 months	- Partners from different cuisine types, different price points and different competitiveness could be matched - More usable at total order level, less at segment level *Next step: add geographical + density + cuisine matching conditions*

None of the methodologies provide results that are reliable enough (partner look alike tbd)

	Smart Growth (Wave 1) - DE		Smart Growth (Wave 2) - DE		Winback UK	
	Incr. orders	Difference with A/B test	Incr. orders	Difference with A/B test	Incr. orders	Difference with A/B test
AB Test *(Ground Truth)*	8.1K	-	2.2K	-	4.0K	-
City Lookalike	22.4K	+14.3K	21.4K	+19.2K	0.0K	-4.0K
Customer Lookalike	3.1K	-5.1K	0	-2.2K	2.2K	-1.8K
Partner Look Alike	To be added					

Proposed next steps

The golden standard to measure causal effects of our campaigns are unbiased A/B tests

Currently we are not able to do this, but these could be enabled by connecting SERP interruptor and OMT to Braze and SFMC/ Hightouch.

Pragmatic Phased Improvement process with continuous alignment

Phase 1: Accuracy Benchmarking

- A/A Testing for each of the evaluation techniques: Run historical simulations to confirm zero-bias during non-testing periods
- Map previous A/B tests to synthetic techniques to quantify the "Bias Gap".

Alignment with DE, NL, UK for multiple initiatives

Approx. 2 weeks (Apr 8)

Phase 2: matching optimization and evaluation technique selection

Optimize current techniques with quantifiable performance measures.
For example:

- *customer and partner matching on city selection (output of city look alike) to limit geographical bias*
- *Customer matching on similar predicted orders in the future*

Alignment with DE, NL, UK for chosen technique per segment per initiative type

Approx. 2 weeks (Apr 22)

Phase 3: Experimentation Logic Upgrade & Automation

- Automated evaluation pipelines with checks on unbiased pre period
- Campaign input based on Google Sheets
- Global Dashboard containing all campaigns from Google Sheet input

tbd

Proposed next steps

The only way to measure the impact of our campaigns accurately is by <u>enabling end to end experimentation</u>

Connecting SERP interruptor and OMT to Braze and SFMC/ Hightouch

Improve Current Techniques

<u>Optimize techniques</u>

Check performance for multiple AB tests (market specific):

- Customer and partner matching on city selection (output of city look alike) to limit geographical bias
- Customer matching on similar predicted orders in the future
- Testing different matching conditions

Approx. 2 weeks (Apr 8)

<u>Recommendations & Alignment</u>

Conclusions and recommendations at segment level:

- Discussing and checks with market analytical teams
- Possible to evaluate Yes/No
- Recommended technique
- Aligned approach across markets

Approx. 2 weeks (Apr 22)

<u>Standardize and scale</u>

- Automated evaluation pipelines
- Campaign input based on Google Sheets
- Global Dashboard containing all campaigns from Google Sheet input

tbd

Smart Growth (Wave 1) - DE

	Total		Early/Engaged		Dormant/Churned/Lapsed	
	Incr. orders	Difference with A/B test	Incr. orders	Difference with A/B test	Incr. orders	Difference with A/B test
AB Test (*Ground Truth*)	8.1K	-	8.1K	-	8.1K	-
City Lookalike V1	8.1K	-	8.1K	-	8.1K	-
Customer Lookalike	8.1K	-	8.1K	-	8.1K	-
Partner Look Alike	8.1K	-	8.1K	-	8.1K	-

Smart Growth Measurement Models

Smart Growth: sending a voucher after ..

	Treatment definition	Control definition	Considerations
AB Test (*Ground Truth*)	Targeting 96k that receive email/push in focus cities	Hold-out group of 11k customers in same cities	No selection bias
City Lookalike	All customers of targeted segment in focus cities with propensity score >X	For each focus city, the cities that were most identical (BSTS) out of 22 other DE cities in Oct-Dec '25	- Jan-Mar '26 are ignored, control cities may have diverged - Overlap and double counting of other campaigns
Customer Lookalike	All customers of targeted segment in focus cities with propensity score >X	For each treatment customer, look alikes (KNN) are found outside the focus cities in Mar '25 to Mar '26	- Jan-Mar '26 are ignored, control cities may have diverged - Overlap and double counting of other campaigns
Partner Look Alike	All partners in focus cities	For each partner in focus city, look alikes are found outside the focus cities in Mar '25 to Mar '26	Local seasonality

Campaign Analysis

Smart Growth (Wave 1) - DE

02.01.2026 - 13.01.2026		Customers	Total iOV	Delta (iOV)
AB Test *(Ground Truth)*	Total	96K	8,1K	
	Early/Engaged	45K	4.3k	
	Dormant/Churned/Lapsed	51.6K	3.8k	
City Lookalike V1 *(Monte Carlo Simulation)*	Early/Engaged	122K	35K	+30.7K
	Dormant/Churned/Lapsed	724.8K	8.9k	+5.1k
City Lookalike V2 *(Locked Audience from AB Test)*	Early/Engaged	36.7K	36	-4.2K
	Dormant/Churned/Lapsed	61.3K	447	-3.4k
Customer Lookalike *(Locked Aucience from AB Test)*	Early/Engaged	21.9K	3.8K	-0.5K
	Dormant/Churned/Lapsed	18K	3.9k	+0.1k

Slide 8

Slide 6

Slide 7

Reconciling Our Measurement Models

Understanding the "Volume Gap" between the A/B Test and Lookalike Models

The Ground Truth: The A/B Test (96.8K Users)

- **Definition:** A "Static Snapshot" of exactly who we sent the voucher to on Launch Day.
- **The Baseline:** Uses a local Hold-out Group (10.7K users) in the same cities.

Methodology 1: Customer Lookalike (1:1 Matching)

- **The Audience**: 39.9K Users (Precision Focus)
- **Why the difference?** We applied a "Rigor Filter" to exclude low-risk users who would have ordered anyway. By focusing only on the High-Risk (35%+) customers, we remove "organic noise."
- **The Logic**: Uses KNN Matching to find a "Twin" for our customers in a different city (e.g., matching a Dortmund user to a Hannover user).
- **The Math**: It calculates the "distance" between two people's behaviors. It doesn't care about "trends" or "time series"; it only cares about individual similarity.

Methodology 2: City Lookalike (Audience-Locked)

- **The Audience**: 98.0K Users (Market Focus)
- **Why the difference?** It is a "Dynamic Query." It captures the 96.8K treated users PLUS ~1.2K users who organically entered the "High-Risk" segment or moved into the city during the campaign.
- **The Logic**: Uses Bayesian Structural Time Series (BSTS) to compare the entire city's daily sales trend against 22 other German cities to predict a "Counterfactual" baseline.
- **The Math**: It calculates correlations over time. It doesn't care about individual "twins"; it only cares about the relationship between city-wide trends.

Reconciling Our Measurement Models

Why does Customer Look Alike find 96% of the uplift in 40% of the audience?

1. The "Voucher Waste" Discovery (The 57k "Organic" Users)

The A/B test (Ground Truth) includes everyone, including 57k users who were "Low-Risk."

- **The Reality**: These 57k users are loyalists. They were going to order anyway. When we send them a voucher, they use it, but they do not drive incremental growth.
- **The Math**: By excluding them with our Rigor Filter (>35% Prob), we are proving that 96% of the campaign's true success came from the "High-Risk" 40k.

2. The "Multiplier" Factor (Eli's 1.5x Correction)

The Customer Lookalike uses Contamination Multiplier (1.5x)

- **Why?** In the Maverick market (Dortmund/Essen), even our "Lookalikes" are influenced by the campaign's presence (advertising, local hype).
- **The Correction**: The 7.8k total includes a +50% "correction" to account for this contamination.
- **The Comparison**: Without the 1.5x multiplier, the Customer Look alike would only report ~5.2k orders. The multiplier is what brings us into alignment with the "Raw Truth" of the A/B test.

3. Signal-to-Noise Ratio (The City Lookalike Failure)

Use the City Lookalike result as your counter-example:

- The City Lookalike V2 used 98k people (the full city volume).
- **The Result**: It only found 447 incremental orders (a 6% accuracy).
- **The Lesson**: This proves that when you include the "Noise" (the other 58k people), the mathematical model breaks. It can't "see" the 8,100 incremental orders because they are buried under the organic behavior of the 58,000 active users.

Reconciling Our Measurement Models

Why did City Lookalike V1 "over-report" by 750,000 customers?

1. The "Open Geography" Trap (Total City vs. Targeted List)

- **The Ground Truth (96.8K):** This is a "Closed List." We only track the specific people who were sent the voucher.
- **City Lookalike - Monte Carlo (846K)**: This model had No Audience Lock. It didn't look at our customer list; instead, it looked at every single person in Dortmund, Essen, and Dresden who was currently in a "High-Risk" segment.
- **The Difference**: This methodology captured 750,000 extra people who never received a voucher. This "Open Market" approach makes it impossible to separate our campaign's impact from general city-wide trends.

2. The "Dormant Data Explosion" (The 724.8K Audience)

- **The Reality**: In a massive market like Germany, there are hundreds of thousands of customers who haven't ordered in years.
- **The Error**: V1 included all 724.8K Dormant/Churned users in its calculation. It "saw" the natural, organic return of a few thousand of these people and incorrectly attributed their orders to our voucher campaign.
- **The Result**: This led to a "Hallucinated Uplift" of +8.9K orders in a segment where the Ground Truth proved only +3.8K were real.

3. The "Loyalist Noise" (The 122K vs. 45K Mismatch)

- **The Logic**: V1 measured 122K Early/Engaged customers (instead of the 45K we actually targeted).
- **The Lesson**: By including an extra 77,000 active loyalists, the model mistook their regular, everyday orders for "Incremental Growth." This is why V1 reported an impossible +30.7K incremental orders—it was simply counting people who were going to order anyway.

Note on Simulation Stability & Methodology Accuracy:
While increasing the number of external model runs (e.g., 100x) and averaging the results is a common technique for reducing "random noise," it does not resolve the Systemic Bias identified in the City Look alike (Monte Carlo) methodology. The underlying Bayesian engine (Hamiltonian Monte Carlo) already executed thousands of internal iterations per run to find a statistically stable mean; therefore, the reported over-estimation (+35K IOV) is not a result of mathematical "jitter," but rather a fundamental failure in Audience Selection.

Campaign Analysis

Smart Growth - Phase 2

19.11.2025 - 30.11.2025		Customers	Total iOV	Delta (iOV)
AB Test *(Ground Truth)*	Total	...	...	
	Early/Engaged	...	...	
	Dormant/Churned/Lapsed	...	...	+30.7K
City Lookalike *(Monte Carlo Simulation)*	Early/Engaged	...	...	+5.1K
	Dormant/Churned/Lapsed	...	...	+5k
City Lookalike *(Locked Aucience from AB Test)*	Early/Engaged	...	...	+982
	Dormant/Churned/Lapsed	...	...	+5k
Customer Lookalike *(Locked Aucience from AB Test)*	Early/Engaged	...	...	+982
	Dormant/Churned/Lapsed	...	...	+5k

Campaign Analysis

Winback -UK

19.11.2025 - 30.11.2025		Customers	Total iOV	Delta (iOV)
AB Test *(Ground Truth)*	Total	...	...	
	Early/Engaged	...	...	
	Dormant/Churned/Lapsed	...	...	+30.7K
City Lookalike *(Monte Carlo Simulation)*	Early/Engaged	...	...	+5.1K
	Dormant/Churned/Lapsed	...	...	+5k
City Lookalike *(Locked Aucience from AB Test)*	Early/Engaged	...	...	+982
	Dormant/Churned/Lapsed	...	...	+5k
Customer Lookalike *(Locked Aucience from AB Test)*	Early/Engaged	...	...	+982
	Dormant/Churned/Lapsed	...	...	+5k

3. Recommendation

Recommendation

Unified Evaluation

When Control group doesn't exist

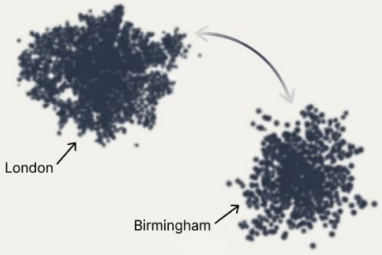
How is the Campaign targeting?

New Prospect

Do we have historical data?

No

Use City Look-alike
OR No Evaluation



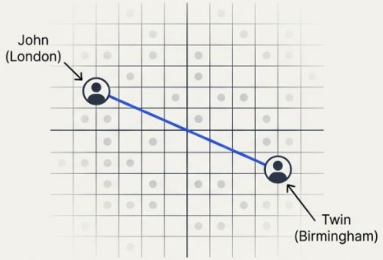
Looking at the crowd.

Existing Customer

Do we have historical data?

YES

ALWAYS use Customer
Look-alike (1:1 Matching).



Looking at the individual.